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Analyzing Depression Severity and Classification: A Comparative Study of Tweets and Survey Responses

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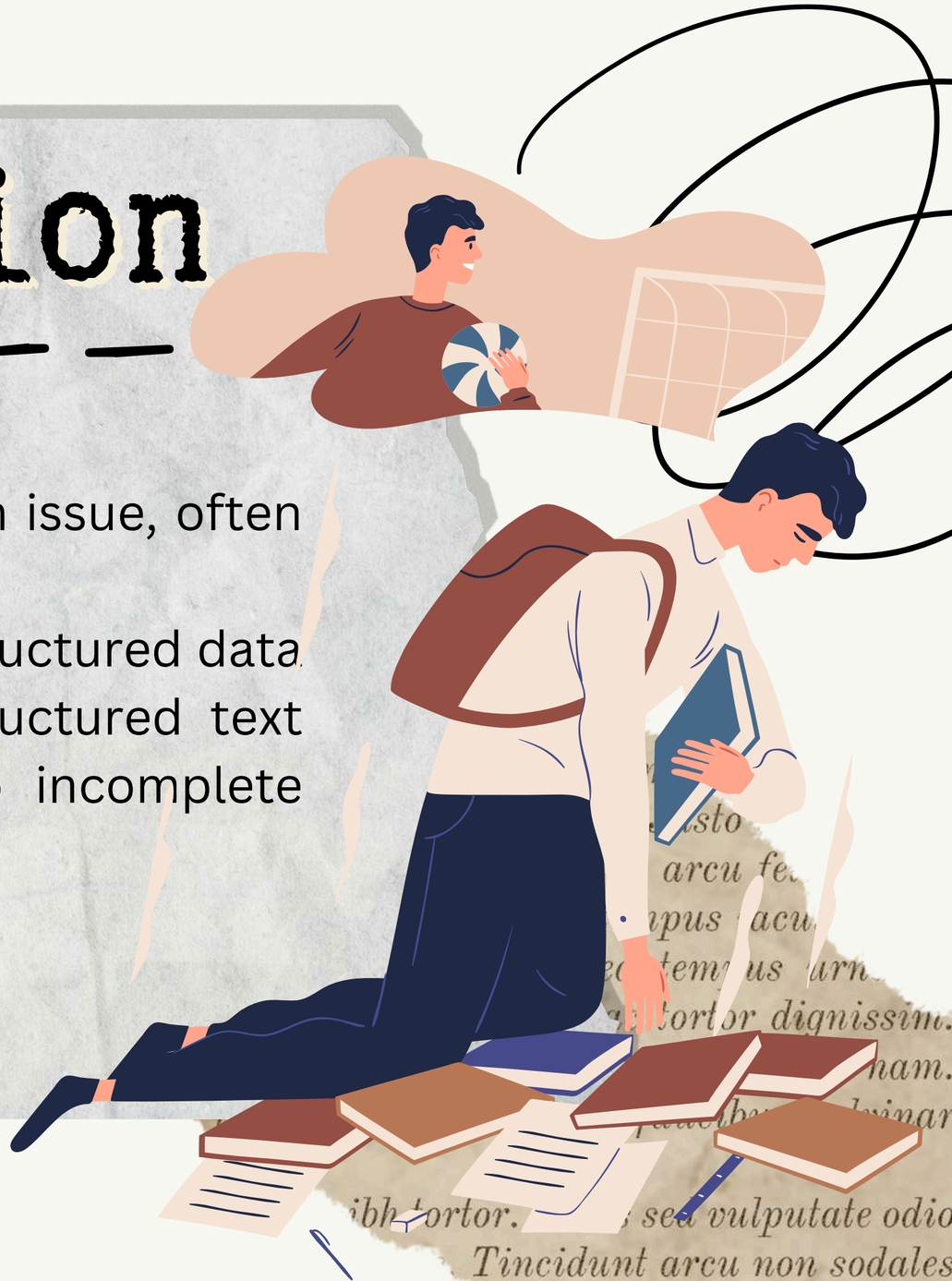
The project

Abstract

Depression is a global mental health concern that often remains undetected until severe stages, emphasizing the need for early detection. This research proposes a hybrid approach combining structured and unstructured data analysis for depression detection. Structured data, such as responses to standardized tools like PHQ-9 and BDI, are analyzed using machine learning models (e.g., Logistic Regression, Decision Trees, SVM) to predict depression risk. Unstructured data, including social media posts, are processed with Natural Language Processing (NLP) techniques to extract linguistic patterns indicative of depressive tendencies. The integration of these methods creates a scalable, efficient system for real-time detection, contributing to early mental health interventions and enhanced well-being.

Introduction

- Depression is a major mental health issue, often undiagnosed due to limited tools.
- Existing systems focus on either structured data (questionnaire responses) or unstructured text (social media analysis), leading to incomplete results.



Problem Statement

Current systems either:

- Focus on clinical questionnaire data, missing behavioral nuances.
- Analyze social media text without incorporating validated clinical scales.
- This results in limited accuracy and scope for early detection.

- Existing depression detection tools often:
  a. Analyze only structured quiz data (e.g., PHQ-9).
  b. Focus solely on social media text using NLP.

These methods lack integration and fail to provide accurate predictions.

Motivation

- Existing depression detection tools often.
- Analyze only structured quiz data (e.g., PHQ-9).
- Focus solely on social media text .
- These methods lack integration and fail to provide accurate predictions.

Objectives

- Research and design a framework integrating structured and unstructured data.
- Explore the use of machine learning for quiz data and NLP for social media analysis.
- Develop a hybrid model combining both approaches for improved accuracy.
- Provide insights into potential implementation challenges and solutions.

Proposed Research Framework

Data Sources:

- Clinical: Questionnaire data like PHQ-9.
- Social: Text from social media posts (tweets, captions, etc.).

Key Steps:

1. Data Collection: Gather diverse, anonymized datasets.
2. Preprocessing: Tokenize text, remove stopwords, normalize scores.
3. Model Design:
   - ML algorithms for structured quiz data.
   - NLP for temporal and semantic text patterns.
4. Hybrid Prediction: Combine insights for final output.

System Design (Conceptual)

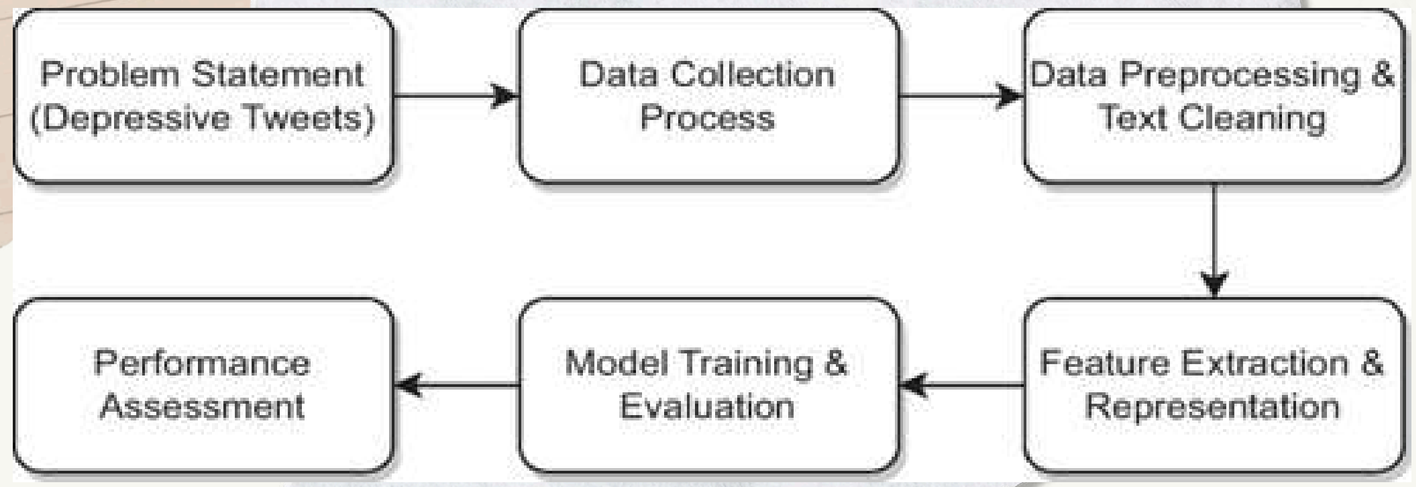
Structured Data Flow:

- Input: Questionnaire scores → ML Model → Preliminary Risk Assessment.

Unstructured Data Flow:

- Input: Social media text → Preprocessing → Sentiment Analysis.

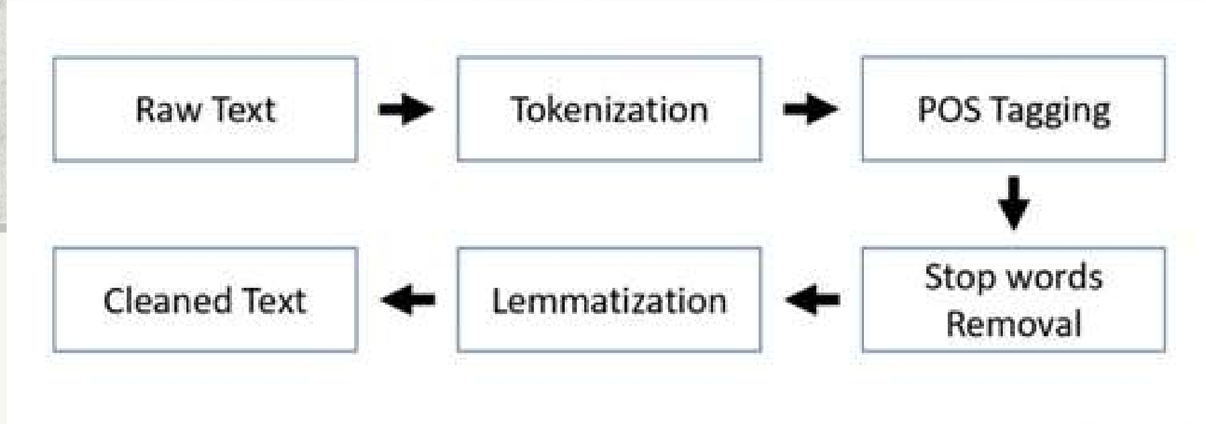
Integration: Hybrid model unifies outputs to improve detection accuracy.



Time plan

Research Challenges

Data Collection:
Ensuring access to ethical, anonymized datasets for clinical and social data.
Feature Extraction:
Optimizing feature engineering for structured and unstructured data simultaneously.
Hybrid Model Design:
Combining two disparate data types in a seamless prediction pipeline.
Validation:
Designing experiments to test accuracy, reliability, and usability of the system.



Expected Outcomes

- A conceptual framework for a hybrid depression detection system.
- Research-driven insights into integrating structured and unstructured data effectively.
- Potential for a scalable implementation in healthcare settings for early intervention.

Future Scope

- Implement and test the framework using real-world datasets.
- Expand the scope to include:
  a. Real-time data collection (e.g., wearable devices).
  b. Multimodal analysis (text, voice, and video).
  c. Integration with healthcare apps for better accessibility and reach.
- Develop a mobile app for live monitoring and consultation support.

Conclusion

- The hybrid system provides a comprehensive and accurate tool for depression detection.
- By combining clinical and behavioral data, it bridges the gap in current detection methods.
- The system supports early intervention, contributing to better mental health outcomes.

References

- M. S. B. Kumar, P. S. Kumar, & S. K. Singh. (2021). "Depression Detection Using Machine Learning: A Survey." International Journal of Advanced Engineering and Management(IJAEM),

This paper reviews various machine learning techniques used for depression detection, focusing on how these models trained on both structured and unstructured data.

- M. B. Gritti, M. P. Ruiz, A. F. Guizelini, & C. T. Igel. (2020). "A Depression Detection System Using Machine Learning Algorithms and Natural Language Processing Techniques."

This paper discusses the use of machine learning and NLP for depression detection, particularly through social media posts.

Thank You

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